

Wilson Costa

Senior Backend Engineer | Distributed Systems | Cloud-Native Platforms
Lisbon, Portugal (UTC+0) | Remote (EU/Worldwide)

Portfolio: wilsoncosta-portfolio.vercel.app | **Github:** github.com/WilsonRCosta

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Summary

Senior Backend Engineer with 5 years specializing in high-throughput distributed systems and performance-critical data pipelines. Proven track record transforming legacy batch workflows into real-time streaming architectures and optimizing production systems from minutes to milliseconds. Experience working with cross-functional and geographically distributed teams; fluent in English (C2).

Key Achievements

- Reduced critical data import latency from 20 minutes to 15 seconds by re-architecting pipelines with optimized SQL and in-memory graph traversal.
- Designed PostgreSQL partitioning and indexing strategies achieving ~500 ms queries on 10M+ row tables for core business workflows.
- Implemented near real-time Kafka-based migration flows processing thousands of multi-megabyte events, eliminating legacy nightly batch bottlenecks.
- Led a multifunctional team as Scrum Master while delivering core platform features, improving sprint predictability and delivery consistency.

Professional Experience

Senior Backend Engineer | Scrum Master

BMW - Critical Techworks (Devoteam)

February 2023 – November 2025

Platform serving internal systems processing millions of records daily.

- Senior backend engineer working on a distributed microservices platform built with Quarkus and PostgreSQL, deployed on Kubernetes with CI/CD pipelines and infrastructure automation (Terraform/Terragrunt on Azure).
- Served as Scrum Master in a team of 7 engineers, facilitating sprint ceremonies and coordinating cross-functional delivery in hybrid and remote environments while maintaining full engineering responsibilities.

- Led the design of a Kafka-based async migration pipeline processing thousands of ~2MB events in near real-time, replacing legacy batch workflows and eliminating processing bottlenecks.
- Re-architected import pipelines using optimized SQL strategies and in-memory graph traversal, achieving an 80× performance improvement (20 minutes to 15 seconds per event) and reducing full migration timelines from multiple days to a few hours.
- Designed advanced PostgreSQL partitioning and indexing strategies, achieving ~500ms queries on 10M+ row tables.
- Implemented complex scheduled processing and logical selection engines enabling millions of vehicle variant assignments to be filtered without combinatorial explosion.
- Improved reliability of multi-service deployments by redesigning CI/CD workflows and production rollout strategies reducing build times by 20%.

Fullstack Engineer

BMW - Critical Techworks (Devoteam)

October 2021 - February 2023

- Delivered fullstack features for a modular dashboard ecosystem supporting real-time widgets and third-party integrations.
- Designed and built APIs using Quarkus and NestJS with GraphQL, PostgreSQL, and CosmosDB.
- Designed and delivered modular widgets used by 1000+ active users.
- Maintained Kubernetes deployments using Helmfile, Jenkins, and ArgoCD pipelines.
- Led Angular migration from v12 to v14 with internal component library modernization.
- Designed high-performance APIs consistently achieving sub-second response times.
- Implemented CMS integration via Strapi for dynamic content delivery.

Fullstack Engineer

Ellucian - Morphis Tech (Devoteam)

March 2021 - September 2021

- Modernized legacy Oracle Forms systems into web-based platforms.
- Built AngularJS frontends and Groovy/Grails backend APIs.
- Refactored business-critical logic for reliability and performance.
- Rebuilt data workflows using JSON APIs for improved frontend/backend integration.
- Optimized database access and API performance in Oracle SQL systems.

Education

Instituto Superior de Engenharia, Lisbon (ISEL)

Bachelor's Degree in Informatics and Computer Engineering

2016 - 2020

Focus: Programming, Web Development, Information Systems, Cloud Computing, Security, Networking

Master's coursework completed (1 year)

Focus: Software Engineering, Distributed Systems, Cybersecurity

Skills

Programming Languages: Java, SQL, TypeScript, JavaScript, Kotlin

Backend & Frameworks: Quarkus, Spring Boot, NodeJS, Kafka, GraphQL, NestJS

Cloud & DevOps: Kubernetes, Docker, Azure, AWS, GitHub Actions, Terraform, Helm, Jenkins, Grafana

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch, CosmosDB, Oracle SQL

Frontend: Angular, React

System Design: Microservices, distributed systems, CI/CD, performance optimization, API design, event-driven architecture, scalability, observability, monitoring

Spoken Languages: Portuguese (native), English (fluent - C2), Spanish (basic)

Certifications

- Professional Scrum Master – Scrum.org | 2024
- AWS Certified Cloud Practitioner – Amazon Web Services | 2022
- Associate Cloud Engineer – Google Cloud | 2021

Projects

Shining Code – E-commerce Platform

Fullstack marketplace platform with product management features.

Live demo: shining-code.onrender.com | **Repository:** github.com/WilsonRCosta/shining-code

Tech: NodeJS/Express, MongoDB, React

- Designed and implemented a Node.js/Express backend, building RESTful APIs for product catalog, cart, authentication, and admin operations.
- Engineered a JWT authentication system with access/refresh token rotation and middleware-based authorization, improving session reliability and API security.
- Implemented media storage in MongoDB GridFS and designed file upload pipelines using Multer, enabling scalable product image handling and consistent retrieval APIs.

- Implemented server-side filtering and query optimization for product endpoints, reducing client complexity and improving API response efficiency.
- Refactored backend architecture during dependency upgrades (Express, Mongoose), improving maintainability and long-term stability.
- Added Swagger API documentation and deployment configuration, streamlining onboarding and production readiness.

Travel Planner App

Fullstack platform to find and reserve restaurants and hotels.

API: github.com/WilsonRCosta/travel-planner-api | **UI:** github.com/WilsonRCosta/travel-planner-app

Tech: Spring Boot, PostgreSQL, Firebase, Google Cloud, Kubernetes, Angular

Backend Description:

- Developed travel management REST API using Java Spring Boot 3.0, PostgreSQL, Firebase, and JWT authentication, integrating TripAdvisor API for real-time hotel and restaurant data with caching optimization.
- Built a comprehensive reservation system with admin approval workflows, user favorites, review management, and Q&A functionality, implementing role-based access control for secure multi-user operations.
- Integrated Google Cloud services (Firebase, Translation API) for authentication, data management, and multi-language support, deploying containerized application using Docker and Kubernetes with complete Swagger/OpenAPI documentation for 20+ REST endpoints